UNIT 6 · ARS/NET PATTERN

MCQ Bank — 140 Questions with Answers

Dynamic soil properties, stress–strain and theories of soil failure (Q1–Q35)

Q1. Soil dynamics is the study of soil behaviour in relation to:

(a) Soil classification
(b) Forces applied by tillage tools and traction devices
(c) Soil chemistry
(d) Rainfall

Q2. The foundational reference 'Soil Dynamics in Tillage and Traction' (USDA Handbook 316) was authored by:

(a) Kepner & Bainer
(b) Gill & Vanden Berg
(c) Goryachkin
(d) Bekker

Q3. A 'dynamic' soil property differs from a static one because it is measured:

(a) In a laboratory only
(b) While the soil is being deformed/failed
(c) After drying
(d) By remote sensing

Q4. The Mohr–Coulomb shear strength equation is:

(a) $\tau = c - \sigma \tan\varphi$
(b) $\tau = c + \sigma \tan\varphi$
(c) $\tau = \sigma - c$
(d) $\tau = c \cdot \sigma$

Q5. In $\tau = c + \sigma \tan\varphi$, the term c represents:

(a) Angle of friction
(b) Cohesion (shear strength at zero normal stress)
(c) Normal stress
(d) Density

Q6. The angle φ is the:

(a) Slope angle
(b) Angle of internal friction
(c) Rake angle
(d) Cone apex

Q7. On a τ–σ plot, cohesion c is the:

(a) Slope
(b) Intercept on the τ-axis
(c) Area
(d) X-intercept

Q8. Mohr's theory says failure occurs when the stress circle:

(a) Is a point
(b) Becomes tangent to the failure envelope
(c) Has zero radius
(d) Lies below the axis

Q9. For a dry cohesionless sand, cohesion c is:

(a) Very high
(b) Zero
(c) Negative
(d) Infinite

Q10. In a saturated clay ($\varphi \approx 0$), shear strength τ equals:

(a) $\sigma \tan\varphi$
(b) c (independent of normal stress)
(c) Zero
(d) σ

Q11. Adding weight to a drive wheel improves traction in sand because thrust depends on:

(a) Cohesion only
(b) Normal load × tanφ
(c) Moisture
(d) Colour

Q12. In very wet clay, adding weight to the wheel mainly causes:

(a) More traction
(b) Sinkage and compaction with little traction gain
(c) Cooling
(d) Less slip

Q13. For a plate of area A shearing soil, maximum force F =

(a) A/c
(b) Ac + W tanφ
(c) W/A
(d) c − W

Q14. Plotting maximum shear force F versus normal force W gives a line whose slope is:

(a) c
(b) tanφ
(c) A
(d) W

Q15. The intercept of that F–W line equals:

(a) tanφ
(b) A·c
(c) W
(d) φ

Q16. The contact 'footprint' area of a rubber tyre is approximated as:

(a) b·l
(b) 0.78 b·l
(c) $\pi b^2/4$
(d) 2bl

Q17. Cohesion arises mainly from:

(a) Inter-particle attraction and moisture films
(b) Wheel load
(c) Air
(d) Temperature

Q18. Soil sticking to a tillage surface is governed by:

(a) Cohesion only
(b) Adhesion and soil–metal friction
(c) Internal friction only
(d) Cone index

Q19. Cone index is:

(a) A soil colour
(b) Force per unit base area to push a cone into soil at a steady rate
(c) Slip percentage
(d) Moisture ratio

Q20. The cone penetrometer is standardised in:

(a) ASAE S313.2
(b) ISO 500
(c) IS 4468
(d) SAE J833

Q21. The standard soil-cone apex angle is:

(a) 15°
(b) 30°
(c) 60°
(d) 90°

Q22. For tyre sinkage < 7.5 cm, the best-correlating cone index is the average over:

(a) 0–5 cm
(b) 0–15 cm (0–6 in)
(c) 0–30 cm
(d) 0–60 cm

Q23. Cone index for traction prediction should normally be measured:

(a) After heavy traffic
(b) Before the soil is subjected to wheel traffic
(c) After rain
(d) At night

Q24. In freshly-tilled compactible soil, after-traffic cone index compared with before-traffic is:

(a) Lower
(b) Same
(c) Several times higher
(d) Zero

Q25. Dense, dry soil shows a stress–strain curve that is:

(a) Purely linear
(b) Brittle — a peak then a lower residual
(c) Flat
(d) Ever-rising with no peak

Q26. Loose, wet soil tends to:

(a) Shatter cleanly
(b) Flow/smear (plastic, work-hardening)
(c) Stay rigid
(d) Explode

Q27. Best pulverization at least draft is obtained by tilling soil near its:

(a) Liquid limit
(b) Friable moisture (near plastic limit)
(c) Saturation
(d) Oven-dry state

Q28. Tilling wet plastic soil causes:

(a) Fine tilth
(b) Smearing, clods and compaction
(c) Lower draft
(d) No effect

Q29. Atterberg limits describe soil:

(a) Colour
(b) Consistency states (liquid, plastic, shrinkage limits)
(c) Strength directly
(d) Cone index

Q30. The plasticity index equals:

(a) LL − PL
(b) PL − SL
(c) LL + PL
(d) LL × PL

Q31. The Bekker bevameter measures:

(a) c only
(b) Pressure–sinkage parameters k_c, k_φ, n
(c) Cone index
(d) Moisture

Q32. Cohesion and internal friction can be obtained directly from a:

(a) Cone penetrometer
(b) Direct-shear / triaxial test
(c) Bomb calorimeter
(d) Patternator

Q33. Under a wheel, soil often fails by:

(a) Shear only
(b) Compression (volume change / densification)
(c) Melting
(d) Evaporation

Q34. Ahead of a narrow tine, brittle soil typically fails by:

(a) Flow
(b) Crescent/shear rupture into blocks
(c) Compression only
(d) No failure

Q35. The universal earthmoving (Reece) equation expresses tool draft as a sum of terms for:

(a) Only weight
(b) Soil weight, cohesion, surcharge and adhesion
(c) Fuel and speed
(d) Slip and sinkage

Answers

Q1 — (b) Forces applied by tillage tools and traction devices

Q2 — (b) Gill & Vanden Berg

Q3 — (b) While the soil is being deformed/failed

Q4 — (b) $\tau = c + \sigma \tan\varphi$ — *Liljedahl p. 241 — $F = Ac + W\tan\varphi \Rightarrow \tau = c + \sigma\tan\varphi$.*

Q5 — (b) Cohesion (shear strength at zero normal stress)

Q6 — (b) Angle of internal friction

Q7 — (b) Intercept on the τ-axis — *c is the τ-intercept, φ the slope.*

Q8 — (b) Becomes tangent to the failure envelope — *Mohr circle tangent to the envelope = failure.*

Q9 — (b) Zero — *Sand $c \approx 0$ — weight-driven thrust.*

Q10 — (b) c (independent of normal stress) — *Saturated clay $\varphi \approx 0$ so $\tau = c$ — weight-independent.*

Q11 — (b) Normal load × tanφ

Q12 — (b) Sinkage and compaction with little traction gain — *Liljedahl notes $\varphi \to 0$ as soil gets very wet; ballast helps in sand, not in wet clay.*

Q13 — (b) Ac + W tanφ — *Plate/lug derivation, Liljedahl pp. 240–242.*
Q14 — (b) tanφ — *Fig. 10-2 — the F–W plot gives slope tanφ.*
Q15 — (b) A·c — *Fig. 10-2 — intercept A·c.*
Q16 — (b) 0.78 b·l — *Fig. 10-3 — tyre footprint A ≈ 0.78 b·l.*
Q17 — (a) Inter-particle attraction and moisture films
Q18 — (b) Adhesion and soil–metal friction
Q19 — (b) Force per unit base area to push a cone into soil at a steady rate — *Liljedahl p. 251 — cone-index definition.*
Q20 — (a) ASAE S313.2
Q21 — (b) 30°
Q22 — (b) 0–15 cm (0–6 in) — *p. 251 — 0–15 cm average for sinkage < 7.5 cm.*
Q23 — (b) Before the soil is subjected to wheel traffic — *p. 252 — measure before traffic.*
Q24 — (c) Several times higher — *p. 252 — compactible soils gain strength under traffic.*
Q25 — (b) Brittle — a peak then a lower residual — *Brittle vs plastic stress–strain — the friable-moisture tillage rule follows.*
Q26 — (b) Flow/smear (plastic, work-hardening)
Q27 — (b) Friable moisture (near plastic limit) — *Least draft and best pulverisation near the plastic limit.*
Q28 — (b) Smearing, clods and compaction
Q29 — (b) Consistency states (liquid, plastic, shrinkage limits)
Q30 — (a) LL - PL — *PI = LL - PL.*
Q31 — (b) Pressure–sinkage parameters k_c, k_φ, n
Q32 — (b) Direct-shear / triaxial test
Q33 — (b) Compression (volume change / densification)
Q34 — (b) Crescent/shear rupture into blocks
Q35 — (b) Soil weight, cohesion, surcharge and adhesion

Mechanics, design and performance of tillage tools (Q36–Q70)

Q36. Fundamentally, a tillage tool acts on soil as a moving:

(a) Roller
(b) Inclined plane / wedge applying compressive stress
(c) Magnet
(d) Heater

Q37. In an uncemented soil, the compressive stress from a tool results in:

(a) Melting
(b) A shearing (failure) action
(c) Evaporation
(d) No change

Q38. Compared with shearing a solid, soil shear reaction:

(a) Is confined to one plane
(b) Extends some distance either side of the shear plane
(c) Is zero
(d) Is instant

Q39. The master geometric variable of a soil-working tool is the:

(a) Colour
(b) Rake (approach) angle
(c) Weight
(d) Serial number

Q40. As rake angle increases well beyond the optimum, draft:

(a) Falls
(b) Rises steeply
(c) Stays constant
(d) Becomes zero

Q41. A narrow tine fails soil ahead of it by:

(a) Melting
(b) A forward/upward crescent shear
(c) Compression only
(d) Boiling

Q42. Below the CRITICAL DEPTH, a narrow tine mainly causes:

(a) Crescent failure
(b) Lateral compaction with little useful loosening
(c) Inversion
(d) Pulverisation

Q43. Working a narrow tine too deep is inefficient because:

(a) Draft rises without useful loosening below the critical depth
(b) It is illegal
(c) Soil gets wetter
(d) Speed drops

Q44. A wide blade's soil resistance is analysed like a:

(a) Beam
(b) Retaining wall (passive earth pressure)
(c) Spring
(d) Pulley

Q45. The universal earthmoving (Reece) equation sums draft terms for:

(a) Fuel and speed
(b) Soil weight, cohesion, surcharge and adhesion
(c) Slip and sinkage
(d) Noise

Q46. Draft of a tillage tool increases with all EXCEPT:

(a) Greater depth and width
(b) Higher soil strength
(c) Lower rake angle within limits
(d) Higher speed

Q47. The dynamic (speed) component of tillage draft varies with:

(a) v
(b) v^2
(c) $\sqrt{v}$
(d) $1/v$

Q48. Unit (specific) draft is draft per unit:

(a) Weight
(b) Furrow cross-sectional area
(c) Time
(d) Wheel

Q49. Nichols found plastic-soil shear force is maximum near the:

(a) Upper plastic (liquid) limit
(b) Lower plastic limit
(c) Oven-dry state
(d) Saturation

Q50. At the upper plastic limit the shear value falls to:

(a) Maximum
(b) Zero
(c) Unchanged
(d) Infinity

Q51. Soil–metal sliding friction μ is roughly independent of clay (colloidal) content above about:

(a) C = 5%
(b) C = 32%
(c) C = 60%
(d) C = 90%

Q52. Highly polished steel reduced μ by 5–20% for:

(a) Sand
(b) High-colloid (clay) soils
(c) Gravel
(d) All equally

Q53. Both the LIFE and the DRAFT of a tillage tool depend strongly on:

(a) Paint colour
(b) Soil–metal friction
(c) Tyre pressure
(d) Engine oil

Q54. Soil sticking to the tool surface (poor scouring) is due to:

(a) Cohesion only
(b) Adhesion, worst in wet high-clay soils
(c) Internal friction only
(d) Cone index

Q55. A remedy for poor scouring in sticky soils is:

(a) Rougher surface
(b) Polished/coated (enamel, PTFE) surface
(c) More weight
(d) Slower speed only

Q56. Wear/abrasion of a share primarily causes:

(a) Loss of suction and higher draft
(b) Lower draft
(c) Better inversion
(d) More suction

Q57. The modern trend in shares is toward:

(a) Welded permanent shares
(b) Disposable/replaceable shares
(c) Wooden shares
(d) No shares

Q58. Down suction of a mouldboard bottom is the:

(a) Vertical clearance dipping the share point below the landside plane
(b) Lateral clearance
(c) Fuel suction
(d) Air gap

Q59. Side suction helps the plough maintain:

(a) Speed
(b) Width of cut
(c) Depth only
(d) Colour

Q60. Typical down/side suction values are about:

(a) 1/8–3/8 inch
(b) 2–3 inch
(c) 6 inch
(d) Zero

Q61. Insufficient suction makes it difficult to maintain:

(a) Desired depth and width of cut
(b) Engine speed
(c) Fuel level
(d) Steering

Q62. The general-purpose mouldboard bottom is characterised by:

(a) No curvature
(b) Moderate curvature for good inversion at moderate speed
(c) Flat plate
(d) Vertical face

Q63. A slat (open) mouldboard is used for:

(a) Dry sand
(b) Sticky soils that will not scour
(c) Rock
(d) Puddling

Q64. Disc ploughs penetrate mainly by:

(a) Suction
(b) Added weight (no suction)
(c) Adhesion
(d) Vibration

Q65. Material hardness of soil-engaging parts is commonly specified by the:

(a) Rockwell C only
(b) Brinell hardness number
(c) Cone index
(d) MWD

Q66. The size-independent measure to compare tillage-tool energy use is:

(a) Total draft
(b) Specific draft / specific energy (kWh/ha)
(c) Mass
(d) Width

Q67. Quality of pulverization is quantified by:

(a) Clod Mean Weight Diameter
(b) VMD
(c) Slip
(d) Cone apex

Q68. Controlled, repeatable evaluation of tillage tools is done in a:

(a) Wind tunnel
(b) Soil bin
(c) Patternator
(d) Calorimeter

Q69. Soil-bin tool forces are measured with:

(a) Thermometers
(b) Strain-gauge (extended-octagonal-ring) dynamometers
(c) Hydrometers
(d) Anemometers

Q70. Tillage-tool draft can be predicted from soil strength using:

(a) Mohr–Coulomb / ASAE D497 / Reece models
(b) Ohm's law
(c) Bernoulli
(d) Nusselt number

Answers

Q36 — (b) Inclined plane / wedge applying compressive stress — *Kepner §7.6, p. 143 — tools as inclined planes/wedges causing shear.*

Q37 — (b) A shearing (failure) action

Q38 — (b) Extends some distance either side of the shear plane — *p. 143 — the shear reaction spreads either side of the plane.*

Q39 — (b) Rake (approach) angle

Q40 — (b) Rises steeply

Q41 — (b) A forward/upward crescent shear

Q42 — (b) Lateral compaction with little useful loosening — *Critical-depth concept (Gill & Vanden Berg).*

Q43 — (a) Draft rises without useful loosening below the critical depth — *Below the critical depth narrow-tine soil compacts laterally instead of rupturing to the surface.*

Q44 — (b) Retaining wall (passive earth pressure)

Q45 — (b) Soil weight, cohesion, surcharge and adhesion

Q46 — (c) Lower rake angle within limits — *A lower rake angle (within limits) REDUCES draft — the exception.*

Q47 — (b) v^2

Q48 — (b) Furrow cross-sectional area

Q49 — (b) Lower plastic limit — *Nichols' data in Kepner pp. 143–144 — shear peaks at the lower plastic limit.*

Q50 — (b) Zero — *Zero at the upper (liquid) plastic limit.*

Q51 — (b) C = 32% — *μ independent of colloid content above C ≈ 32%.*

Q52 — (b) High-colloid (clay) soils — *Polishing cuts μ 5–20% for high-clay soils, not sand.*

Q53 — (b) Soil–metal friction — *Kepner §7.5 — friction sets tool life and draft.*

Q54 — (b) Adhesion, worst in wet high-clay soils

Q55 — (b) Polished/coated (enamel, PTFE) surface

Q56 — (a) Loss of suction and higher draft — *Wear loses suction and raises draft.*

Q57 — (b) Disposable/replaceable shares — *Kepner p. 145 — trend to disposable shares.*

Q58 — (a) Vertical clearance dipping the share point below the landside plane — *Kepner p. 142 — down suction is vertical clearance.*

Q59 — (b) Width of cut

Q60 — (a) 1/8–3/8 inch — *Kepner p. 142 — ⅛–⅜ in.*

Q61 — (a) Desired depth and width of cut — *Without suction, depth/width cannot be held.*

Q62 — (b) Moderate curvature for good inversion at moderate speed

Q63 — (b) Sticky soils that will not scour

Q64 — (b) Added weight (no suction)

Q65 — (b) Brinell hardness number

Q66 — (b) Specific draft / specific energy (kWh/ha)

Q67 — (a) Clod Mean Weight Diameter

Q68 — (b) Soil bin

Q69 — (b) Strain-gauge (extended-octagonal-ring) dynamometers

Q70 — (a) Mohr–Coulomb / ASAE D497 / Reece models

Traction devices, tyres, traction theories, slip and sinkage (Q71–Q105)

Q71. Traction is defined as:

(a) The braking force
(b) The driving force developed by a wheel/track/traction device
(c) Rolling resistance
(d) Tyre pressure

Q72. Tractive efficiency (TE) is the ratio of:

(a) Pull to weight
(b) Output power to input power of the traction device
(c) Slip to sinkage
(d) Torque to speed

Q73. Net traction coefficient μ equals:

(a) Gross force/torque
(b) Net pull ÷ dynamic normal load
(c) Slip%
(d) Rolling radius

Q74. Motion-resistance ratio ρ is:

(a) MR ÷ normal load
(b) Pull ÷ MR
(c) Slip
(d) Sinkage/width

Q75. Net and gross traction coefficients are related by:

(a) $\mu = \mu_g + \rho$
(b) $\mu = \mu_g - \rho$
(c) $\mu = \mu_g \times \rho$
(d) $\mu = \rho - \mu_g$

Q76. Net pull H of a driving wheel equals:

(a) F + MR
(b) F − MR (gross traction minus motion resistance)
(c) W − F
(d) T/r

Q77. Wheel slip is defined as S =:

(a) V_t/V_a
(b) $1 - V_a/V_t$
(c) $V_a - V_t$
(d) $V_a/V_t + 1$

Q78. In the slip formula, V_t is the:

(a) Actual travel speed
(b) Theoretical speed rω on a hard surface
(c) Wind speed
(d) PTO speed

Q79. A driving wheel develops thrust only when it:

(a) Is perfectly rigid
(b) Slips (some travel reduction)
(c) Stops
(d) Is towed

Q80. Rolling radius is standardised in:

(a) ASAE S296.2
(b) ASAE S313.2
(c) ISO 500
(d) SAE J833

Q81. Typical working slip on firm soil is about:

(a) 0%
(b) 8–15%
(c) 40%
(d) 80%

Q82. A towed (unpowered) wheel has:

(a) Maximum torque
(b) Zero axle torque
(c) Zero load
(d) Maximum pull

Q83. For a self-propelled wheel condition:

(a) Pull is zero, torque overcomes motion resistance
(b) Torque is zero
(c) Slip is 100%
(d) Load is zero

Q84. Weight transfer under drawbar pull on a 2WD tractor:

(a) Reduces rear traction
(b) Increases rear-wheel reaction and maximum pull
(c) Has no effect
(d) Lifts the rear

Q85. Increasing rear-wheel reaction increases traction only if:

(a) The soil is strong enough and sinkage does not limit it
(b) Always
(c) Speed is high
(d) Tyre is bald

Q86. Maximum soil thrust of a device is H =

(a) Ac + W tanφ
(b) A/c
(c) W/A
(d) c - W

Q87. In dry sand (c≈0), traction is improved by:

(a) Adding weight (thrust ∝ load)
(b) Reducing weight
(c) Wetting
(d) Nothing

Q88. In saturated clay (φ≈0), adding ballast mainly causes:

(a) More traction
(b) Sinkage/compaction with little gain
(c) Less slip
(d) Higher TE

Q89. The dimensionless wheel numeric (cone-index number) is:

(a) Cn = C·b·d/W
(b) Cn = W/Cbd
(c) Cn = C/W
(d) Cn = bd/C

Q90. In Cn = Cbd/W, C is the:

(a) Cohesion
(b) Cone index (soil strength)
(c) Circumference
(d) Camber

Q91. A higher Cn value indicates:

(a) Weaker soil
(b) Stronger soil or lighter load → better traction
(c) More slip
(d) More sinkage

Q92. The Wismer–Luth net-pull relation is H/W =

(a) 0.75(1 - e^(-0.3CnS)) - (1/Cn + 0.04)
(b) CnS
(c) 0.75CnS
(d) 1 - S

Q93. The term (1/Cn + 0.04) in that equation represents the:

(a) Gross traction
(b) Motion-resistance ratio
(c) Slip
(d) Sinkage

Q94. The gross traction coefficient in the Wismer–Luth model is:

(a) 0.75(1 - e^(-0.3CnS))
(b) 1/Cn
(c) 0.04
(d) CnS

Q95. Tractive efficiency with slip:

(a) Rises indefinitely
(b) Peaks at low slip then falls
(c) Is constant
(d) Is zero at low slip

Q96. Maximum TE occurs at LOWER slip for:

(a) Small Cn
(b) Large Cn (strong soil/light load)
(c) Wet clay
(d) High inflation

Q97. For large drawbar pull, the design slip is chosen:

(a) At the TE peak
(b) To the right of the TE peak
(c) At zero slip
(d) At 100% slip

Q98. In the standard design example, Cn = 30 gives approximately:

(a) TE 0.72, H/W 0.51, slip 0.16 (b) TE 0.30

(c) TE 0.95 (d) TE 0.10

Q99. Sinkage under contact pressure follows Bekker's:

(a) $p = (k_c/b + k_\varphi)\cdot z^n$ (b) $p = z/b$

(c) $p = c + \sigma\tan\varphi$ (d) $p = \pi DN/60$

Q100. Pressure–sinkage parameters are obtained with a:

(a) Cone penetrometer only (b) Bevameter (plate-sinkage)

(c) Patternator (d) Dynamometer

Q101. In a tyre size code 13.6–38, the number 38 is the:

(a) Section width (b) Rim (wheel) diameter in inches

(c) Ply rating (d) Lug angle

Q102. The general-purpose agricultural tyre tread class is:

(a) R-1 (b) R-3

(c) R-4 (d) F-2

Q103. Very deep lugs for wetland/cane-rice tyres denote class:

(a) R-1 (b) R-2

(c) R-3 (d) R-4

Q104. Compared with bias-ply, radial tractor tyres give:

(a) Smaller footprint (b) Larger footprint, less slip, better TE and wear

(c) More rolling resistance (d) No difference

Q105. Most soil compaction from a wheel occurs during the:

(a) First pass (b) Tenth pass

(c) Turning (d) Braking

Answers

Q71 — (b) The driving force developed by a wheel/track/traction device

Q72 — (b) Output power to input power of the traction device — *Liljedahl p. 243 — TE = output/input power.*

Q73 — (b) Net pull ÷ dynamic normal load

Q74 — (a) MR ÷ normal load

Q75 — (b) $\mu = \mu_g - \rho$ — *$\mu = \mu_g - \rho$.*

Q76 — (b) F - MR (gross traction minus motion resistance) — *$H = F - MR$.*

Q77 — (b) $1 - V_a/V_t$ — *p. 245 — $S = 1 - V_a/V_t$.*

Q78 — (b) Theoretical speed rω on a hard surface — *$V_t = r\omega$, rolling radius per ASAE S296.2.*

Q79 — (b) Slips (some travel reduction) — *A wheel must slip to develop thrust.*

Q80 — (a) ASAE S296.2

Q81 — (b) 8–15%

Q82 — (b) Zero axle torque

Q83 — (a) Pull is zero, torque overcomes motion resistance

Q84 — (b) Increases rear-wheel reaction and maximum pull

Q85 — (a) The soil is strong enough and sinkage does not limit it

Q86 — (a) Ac + W tanφ — *Mohr–Coulomb thrust (Topic 2).*

Q87 — (a) Adding weight (thrust ∝ load) — *Sand $c \approx 0$ — ballast helps.*

Q88 — (b) Sinkage/compaction with little gain — *Wet clay $\varphi \approx 0$ — ballast sinks.*

Q89 — (a) Cn = C·b·d/W — *pp. 248–250 — $Cn = Cbd/W$.*

Q90 — (b) Cone index (soil strength)

Q91 — (b) Stronger soil or lighter load → better traction

Q92 — (a) 0.75(1 - e^(-0.3CnS)) - (1/Cn + 0.04) — *Option (a) is the intended answer. The published Wismer–Luth motion-resistance term is 1.2/Cn + 0.04 (see the note in Topic 8).*

Q93 — (b) Motion-resistance ratio — *ρ ≈ 1.2/Cn + 0.04 in the published form; the source prints 1/Cn + 0.04.*

Q94 — (a) 0.75(1 - e^(-0.3CnS)) — *$\mu_g = 0.75(1 - e^{(-0.3CnS)})$.*

Q95 — (b) Peaks at low slip then falls — *p. 250 — TE peaks at low slip.*

Q96 — (b) Large Cn (strong soil/light load) — *Lower slip for larger Cn.*

Q97 — (b) To the right of the TE peak — *Design slip right of the peak for pull.*

Q98 — (a) TE 0.72, H/W 0.51, slip 0.16 — *Cn = 30 → TE 0.72, H/W 0.51, slip 0.16 (p. 250).*

Q99 — (a) $p = (k_c/b + k_\varphi)\cdot z^n$

Q100 — (b) Bevameter (plate-sinkage)

Q101 — (b) Rim (wheel) diameter in inches

Q102 — (a) R-1

Q103 — (b) R-2

Q104 — (b) Larger footprint, less slip, better TE and wear

Q105 — (a) First pass

Soil compaction — causes, effects and alleviation (Q106–Q140)

Q106. Soil compaction is best described as:

(a) Loss of soil water

(b) Densification — bulk density rises and porosity falls

(c) Erosion

(d) Salinity build-up

Q107. In soil-dynamics terms compaction is soil failure by:

(a) Shear

(b) Compression (volume change)

(c) Tension

(d) Melting

Q108. Under a wheel the soil typically fails by compression, whereas under a narrow tine it fails by:

(a) Compression

(b) Shear

(c) Evaporation

(d) No failure

Q109. The DEPTH to which compaction reaches is governed mainly by:

(a) Contact pressure

(b) Axle load

(c) Tyre colour

(d) Speed

Q110. The SEVERITY of surface compaction is governed mainly by:

(a) Axle load

(b) Contact (ground) pressure — inflation and tyre width

(c) Engine power

(d) Slip

Q111. Soil is MOST compactible when it is:

(a) Oven dry

(b) At/near an optimum (field-capacity) moisture

(c) Saturated

(d) Frozen

Q112. According to Liljedahl's Table 10-3, most of the compaction (71–96% of the 4-pass total) occurs on the:

(a) Fourth pass

(b) First pass

(c) Turning

(d) Braking

Q113. The practical lesson from the first-pass finding is to:

(a) Wheel everywhere
(b) Confine traffic to permanent lanes (controlled traffic)
(c) Inflate tyres more
(d) Add ballast

Q114. A dense sub-surface layer formed at the usual tillage depth is called a:

(a) Crust
(b) Plough (hard) pan
(c) Furrow slice
(d) Mulch

Q115. The curve of bulk density versus moisture at fixed effort is the:

(a) Mohr circle
(b) Proctor curve
(c) Willan line
(d) NDVI curve

Q116. On the Proctor curve, maximum bulk density occurs at:

(a) Zero moisture
(b) An optimum moisture content
(c) Saturation
(d) Wilting point

Q117. Very wet soil resists compaction to maximum density because:

(a) It is strong
(b) Pore water carries the load and the soil shears/smears
(c) It is frozen
(d) No particles present

Q118. Compaction reduces especially the soil's:

(a) Sand fraction
(b) Large air-filled macropores
(c) pH
(d) Colour

Q119. As a result of compaction, water infiltration and hydraulic conductivity:

(a) Increase
(b) Decrease
(c) Stay same
(d) Become zero permanently

Q120. Compaction raises subsequent tillage:

(a) Speed
(b) Draft and fuel use
(c) Quality only
(d) Nothing

Q121. Penetration resistance of compacted soil is measured with a:

(a) Patternator
(b) Cone penetrometer
(c) Hydrometer
(d) Anemometer

Q122. A commonly cited root-limiting bulk density is about:

(a) 0.8 g/cm³
(b) 1.1 g/cm³
(c) 1.6–1.8 g/cm³
(d) 3.0 g/cm³

Q123. A cone index (penetration resistance) that seriously restricts roots is roughly:

(a) >0.2 MPa
(b) >2 MPa
(c) >20 MPa
(d) >200 MPa

Q124. Effects of compaction on the crop include all EXCEPT:

(a) Restricted root growth
(b) Poorer aeration and uptake
(c) Higher yield
(d) Poor emergence

Q125. Firming of soil by a press wheel around the seed is:

(a) Harmful compaction
(b) Beneficial for seed–soil contact and germination
(c) Irrelevant
(d) Illegal

Q126. Which reduces ground pressure and hence compaction?

(a) Narrow, over-inflated tyres
(b) Radial/large/dual tyres and tracks at lower inflation
(c) More axle load
(d) Higher speed

Q127. Controlled-traffic farming works by:

(a) Wheeling the whole field
(b) Confining all traffic to fixed permanent lanes
(c) Random driving
(d) Banning tractors

Q128. Heavy field traffic should especially be avoided when the soil is:

(a) Dry
(b) Wet (near/above field capacity)
(c) Frozen hard
(d) Sandy

Q129. Subsoil compaction from very heavy axle loads is problematic because it is:

(a) Easily reversed by rain
(b) Deep and persistent
(c) Beneficial
(d) Temporary

Q130. A pan is best broken by:

(a) Rolling
(b) Subsoiling/chiselling below the pan
(c) Puddling
(d) More passes

Q131. Subsoiling is most effective (best shattering) when the subsoil is:

(a) Saturated
(b) Dry and brittle
(c) Frozen
(d) Puddled

Q132. To locate the depth of a pan before subsoiling, use a:

(a) Cone-index (penetrometer) profile
(b) Thermometer
(c) pH meter
(d) Tape only

Q133. Building organic matter and using deep-rooted cover crops helps by:

(a) Adding weight
(b) Creating biopores and improving structure/resilience
(c) Raising bulk density
(d) Increasing slip

Q134. Bulk density is measured in the field with a:

(a) Core sampler
(b) Patternator
(c) Bevameter
(d) Tachometer

Q135. The most cost-effective compaction strategy is:

(a) Repeated subsoiling
(b) Prevention (traffic & pressure control)
(c) Deep ploughing every year
(d) Ignoring it

Q136. Puddling in paddy fields is a deliberate:

(a) Loosening
(b) Compaction of the subsurface to reduce percolation
(c) Drying
(d) Levelling only

Q137. Compaction and traction are linked because:

(a) A firmed surface can improve a following wheel's traction
(b) They are unrelated
(c) Compaction removes traction
(d) Traction dries soil

Q138. Over-inflating a tractor tyre tends to:

(a) Reduce surface compaction
(b) Increase contact pressure and surface compaction
(c) Increase footprint
(d) Lower draft

Q139. Compared with prevention, correcting compaction by tillage is:

(a) Cheaper

(b) Energy-intensive and temporary if traffic continues

(c) Permanent

(d) Unnecessary

Q140. Which sequence is correct for managing a compacted field?

(a) Subsoil, then keep wheeling wet

(b) Diagnose pan depth → subsoil dry → then prevent re-compaction

(c) Add ballast and drive more

(d) Do nothing

Answers

Q106 — (b) Densification — bulk density rises and porosity falls

Q107 — (b) Compression (volume change) — *Topic 2 distinction — compression (wheel) vs shear (tine) failure of the SAME soil.*

Q108 — (b) Shear

Q109 — (b) Axle load — *Axle load sets the DEPTH of the compacted zone.*

Q110 — (b) Contact (ground) pressure — inflation and tyre width — *Contact pressure (inflation, tyre width) sets the SEVERITY near the surface — a standard exam pair.*

Q111 — (b) At/near an optimum (field-capacity) moisture

Q112 — (b) First pass — *Liljedahl Table 10-3, p. 257 — first pass = 71–91% (bulk density) and 74–96% (sinkage) of the 4-pass total.*

Q113 — (b) Confine traffic to permanent lanes (controlled traffic) — *Hence controlled traffic.*

Q114 — (b) Plough (hard) pan

Q115 — (b) Proctor curve — *Proctor relation — densest packing at optimum moisture.*

Q116 — (b) An optimum moisture content

Q117 — (b) Pore water carries the load and the soil shears/smears — *Wet soil shears/smears instead of densifying to maximum.*

Q118 — (b) Large air-filled macropores

Q119 — (b) Decrease

Q120 — (b) Draft and fuel use

Q121 — (b) Cone penetrometer

Q122 — (c) 1.6–1.8 g/cm³

Q123 — (b) >2 MPa

Q124 — (c) Higher yield

Q125 — (b) Beneficial for seed–soil contact and germination

Q126 — (b) Radial/large/dual tyres and tracks at lower inflation

Q127 — (b) Confining all traffic to fixed permanent lanes

Q128 — (b) Wet (near/above field capacity)

Q129 — (b) Deep and persistent

Q130 — (b) Subsoiling/chiselling below the pan — *Subsoiling below the pan shatters best when the subsoil is dry/brittle (smears if wet).*

Q131 — (b) Dry and brittle — *Links to the brittle-vs-plastic stress–strain behaviour of Topic 2.*

Q132 — (a) Cone-index (penetrometer) profile

Q133 — (b) Creating biopores and improving structure/resilience

Q134 — (a) Core sampler

Q135 — (b) Prevention (traffic & pressure control)

Q136 — (b) Compaction of the subsurface to reduce percolation

Q137 — (a) A firmed surface can improve a following wheel's traction

Q138 — (b) Increase contact pressure and surface compaction

Q139 — (b) Energy-intensive and temporary if traffic continues

Q140 — (b) Diagnose pan depth → subsoil dry → then prevent re-compaction